

# Davide Boscaini, Ph.D.

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## About

I am a Research Scientist at the **Technologies of Vision** research unit of the **Fondazione Bruno Kessler** in Trento, Italy. My research interests are in 3D perception and understanding, with a focus on object 6D pose estimation and 3D scene segmentation. According to Google Scholar, I have an h-index of 16, an i10-index of 22, and my articles have over 6,500 citations.

Before joining FBK, I received a PhD in Computational Science from the **Università della Svizzera italiana**

in Lugano, Switzerland, in 2017. During my PhD, under the supervision of prof. **Michael Bronstein**, my research focused on extending deep learning techniques to geometric domains such as 3D shapes and graphs, contributing to the birth of a new research direction called **Geometric Deep Learning**. Prior to that, I obtained an M.S. in Mathematics from the **University of Verona**, Italy, in 2013, and a B.S. in Applied Mathematics from the same institution in 2010.

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## Education

Ph.D. in Computational Science

**Università della Svizzera italiana**

Dissertation on “Geometric Deep Learning for Shape Analysis”

Advisor: M.M. Bronstein. Co-advisor: J. Masci

Examiners: J. Schmidhuber, M. Ovsjanikov, P. Vanderghenst, K. Hormann

Sep. 2013 – Sep. 2017

*Lugano, Switzerland*

M.S. in Mathematics

**University of Verona**

Dissertation on “Spectral Methods for Shape Analysis”

Advisor: G. Orlandi. Co-advisor: U. Castellani

Oct. 2010 – Mar. 2013

*Verona, Italy*

B.S. in Applied Mathematics

**University of Verona**

Dissertation on “Existence and multiplicity of the solutions of the Plateau problem”

Advisor: S. Baldo

Sep. 2007 – Oct. 2010

*Verona, Italy*

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## Awards

“Second place” for task-driven affordance grounding

SceneFun3D Challenge 2025

“Best overall method” for 6D detection of unseen objects

BOP Challenge 2024

“Best overall method” for 6D localization of unseen objects

BOP Challenge 2024

“Early-bird winner” of 6D detection of unseen objects

BOP Challenge 2024

“Early-bird winner” of 6D localization of unseen objects

BOP Challenge 2024

“Best method on TUD-L dataset” for the 6D localization of unseen objects

BOP Challenge 2023

Outstanding reviewer

CVPR 2025

Outstanding reviewer

CVPR 2024

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## Publications

**Generative 6D pose estimation via conditional flow matching**

ICIP 2026

A. Hamza, D. Boscaini, W. Li, B. Busam, F. Poiesi

*First application of Conditional Flow Matching to 6D pose estimation of seen objects*

**Action-guided generation of 3D functionality segmentation data**

CVPR-W 2026

J. Corsetti, F. Giuliari, D. Boscaini, P. Hermosilla, A. Pilzer, G. Mei, A. Delitzas, F. Engelmann, F. Poiesi

**High-resolution open-vocabulary object 6D pose estimation**

TPAMI, 2026

J. Corsetti, D. Boscaini, F. Giuliari, C. Oh, A. Cavallaro, F. Poiesi

**CHIP: A multi-sensor dataset for 6D pose estimation of chairs in industrial settings**

BMVC 2025

M. Nardon, M. Mujika, A. González, D. Sedano, J. Rueda, A. Caro, A. Caraffa, F. Poiesi, P.I. Chippendale, D. Boscaini

- Distilling 3D distinctive local descriptors for 6D pose estimation** IROS 2025  
A. Hamza, A. Caraffa, D. Boscaini, F. Poiesi
- AI-driven visual monitoring of industrial assembly tasks** ICIAP 2025  
M. Nardon, S. Messelodi, A. Granata, F. Poiesi, A. Danese, D. Boscaini
- Functionality understanding and segmentation in 3D scenes** CVPR 2025  
J. Corsetti, F. Giuliari, A. Fasoli, D. Boscaini, F. Poiesi  
*Highlight poster (top 3%). Awarded "Second place" for "Task-driven affordance grounding" at the SceneFun3D Challenge 2025*
- 3D part segmentation via geometric aggregation of 2D visual features** WACV 2025  
M. Garosi, R. Tedoldi, D. Boscaini, M. Mancini, N. Sebe, F. Poiesi
- Enabling seamless HRC integration: The AI-PRISM reference architecture** I-ESA 2024  
M.A. Mateo-Casalí, J. Rueda, S.O. Afolaranmi, J.L.M. Lastra, D. Boscaini, L. Moya-Ruiz, F. Fraile
- Towards more effective human-robot collaboration via accurate pose estimation** ERF 2024  
M.A. Mateo-Casalí, L. Moya-Ruiz, A. Caraffa, D. Boscaini, A. Hamza, P. Chippendale, F. Poiesi, F. Fraile
- Wild Berry image dataset collected in Finnish forests and peatlands using drones** ECCV-W 2024  
L. Riz, S. Povoli, A. Caraffa, D. Boscaini, M.L. Mekhalfi, P. Chippendale, M. Turtiainen, B. Partanen, L.S. Ballester, F.B. Noguera, A. Franchi, E. Castelli, G. Piccinini, L. Marchesotti, M.S. Couceiro, F. Poiesi
- FreeZe: Training-free zero-shot 6D pose estimation with geometric and vision foundation models** ECCV 2024  
A. Caraffa, D. Boscaini, A. Hamza, F. Poiesi  
*An enhanced version of this work, FreeZe-v2, is the "early bird winner" in both 6D localization and 6D detection at the BOP Challenge 2024. An early version of this work, PoZe, won the "Best method on TUD-L dataset" award at the BOP Challenge 2023*
- Exploring fine-grained retail product discrimination with zero-shot object classification using Vision-Language Models** RTSI 2024  
A. Tur, A. Conti, C. Beyan, D. Boscaini, R. Larcher, S. Messelodi, F. Poiesi, E. Ricci
- Open-vocabulary object 6D pose estimation** CVPR 2024  
J. Corsetti, D. Boscaini, C. Oh, A. Cavallaro, F. Poiesi
- Tracciamento 3D della palla da punti di vista multipli nella pallavolo** Ital-IA 2024  
L. Riz, S. Povoli, D. Boscaini, S. Messelodi, F. Poiesi
- Detect, Augment, Compose, and Adapt: Four steps for unsupervised domain adaptation in object detection** BMVC 2023  
M.L. Mekhalfi, D. Boscaini, F. Poiesi
- Revisiting Fully Convolutional Geometric Features for object 6D pose estimation** ICCV-W 2023  
J. Corsetti, D. Boscaini, F. Poiesi
- PatchMixer: Rethinking network design to boost generalization for 3D point cloud understanding** IMAVIS, 2023  
D. Boscaini, F. Poiesi  
*Novel network design that is intrinsically effective in generalisation across datasets unseen at training time*
- Supervised tractogram filtering using geometric deep learning** MIA, 2023  
P. Astolfi, R. Verhagen, L. Petit, E. Olivetti, S. Sarubbo, J. Masci, D. Boscaini, P. Avesani
- The MONET dataset: Multimodal drone thermal dataset recorded in rural scenarios** CVPR-W 2023  
L. Riz, A. Caraffa, M. Bortolon, M.L. Mekhalfi, D. Boscaini, A. Moura, J. Antunes, A. Dias, H. Silva, A. Leonidou, C. Constantinides, C. Keleshis, D. Abate, F. Poiesi
- Learning general and distinctive 3D local deep descriptors for point cloud registration** TPAMI, 2023  
F. Poiesi, D. Boscaini  
*State-of-the-art performance for point cloud registration in the transfer learning setting across 3DMatch, ETH, and Kitti datasets*

- SHIELD: Safeguard heritage in endangered looted districts** Ital-IA 2022  
M.L. Mekhalfi, N. Saljoughi, D. Boscaini, F. Poiesi
- Localisation of defects in volumetric CT scans of valuable wood logs** ICPR-W 2020  
D. Boscaini, F. Poiesi, S. Messelodi, A. Younes, D. Grande  
*Selected for oral presentation*
- Joint supervised and self-supervised learning for 3D real-world challenges** ICPR 2020  
A. Alliegro, D. Boscaini, T. Tommasi  
*Selected for oral presentation (4.4% acceptance rate)*
- Distinctive 3D local deep descriptors** ICPR 2020  
F. Poiesi, D. Boscaini
- Shape consistent 2D keypoint estimation under domain shift** ICPR 2020  
L.O. Vasconcelos, M. Mancini, D. Boscaini, S. Rota Bulò, B. Caputo, E. Ricci
- Novel-view human action synthesis** ACCV 2020  
M. Lakhal, D. Boscaini, F. Poiesi, O. Lanz, A. Cavallaro
- Clustered dynamic graph CNN for biometric 3D hand shape recognition** IJCB 2020  
J. Svoboda, P. Astolfi, D. Boscaini, J. Masci, M.M. Bronstein
- Tractogram filtering of anatomically non-plausible fibers with geometric deep learning** MICCAI 2020  
P. Astolfi, R. Verhagen, L. Petit, E. Olivetti, J. Masci, D. Boscaini, P. Avesani
- Self-supervision for 3D real-world challenges** ECCV-W 2020  
A. Alliegro, D. Boscaini, T. Tommasi
- Deciphering interaction fingerprints from protein molecular surfaces** Nature Methods, 2020  
P. Gainza, F. Sverrisson, F. Monti, E. Rodolà, D. Boscaini, M.M. Bronstein, B.E. Correira  
*Advertised on the cover of the Feb 2020 issue of the journal*
- Learning interaction patterns from surface representations of protein structure** NeurIPS-W 2019  
P. Gainza, F. Sverrisson, F. Monti, E. Rodolà, D. Boscaini, M.M. Bronstein, B.E. Correira
- Structured domain adaptation for 3D keypoint estimation** 3DV 2019  
L.O. Vasconcelos, M. Mancini, D. Boscaini, B. Caputo, E. Ricci  
*Oral presentation*
- 3D shape segmentation with geometric deep learning** ICIAP 2019  
D. Boscaini, F. Poiesi  
*Spotlight presentation*
- Geometric deep learning on graphs and manifolds using mixture model CNNs** CVPR 2017  
F. Monti\*, D. Boscaini\*, J. Masci, E. Rodolà, J. Svoboda, M.M. Bronstein  
*Oral presentation (top 0.8%). First unified framework generalizing CNN architectures to non-Euclidean domains such as 3D shapes and graphs. Also available as technical report: arXiv:1611.08402. (\* indicates equal contribution)*
- Geometric deep learning** SIGGRAPH Asia Courses 2016  
J. Masci, E. Rodolà, D. Boscaini, M.M. Bronstein, H. Li
- Learning shape correspondence with anisotropic convolutional neural networks** NeurIPS 2016  
D. Boscaini, J. Masci, E. Rodolà, M.M. Bronstein  
*Presented also as a poster at the 3D Deep Learning Workshop (3DLL) 2016. Also available as technical report: arXiv:1605.06437*
- Anisotropic diffusion descriptors** CGF, 2016  
D. Boscaini, J. Masci, E. Rodolà, M.M. Bronstein, D. Cremers  
*Oral presentation at EUROGRAPHICS 2016*
- Geodesic convolutional neural networks on Riemannian manifolds** ICCV-W 2015  
J. Masci\*, D. Boscaini\*, M.M. Bronstein, P. Vandergheynst  
*Oral presentation at 3DRR 2015. First extension of the popular CNN paradigm to non-Euclidean domains. An early version of this work was published as the technical report arXiv:1501.06297 on January 2015. (\* indicates equal contribution)*

**Learning class-specific descriptors for deformable shapes using localized spectral convolutional networks**

CGF, 2015

D. Boscaini, J. Masci, S. Melzi, M.M. Bronstein, U. Castellani, P. Vanderghelynst  
*Oral presentation at SGP 2015*

**Shape-from-operator: Recovering shapes from intrinsic operators**

CGF, 2015

D. Boscaini, D. Eynard, D. Kourounis, M.M. Bronstein  
*Oral presentation at EUROGRAPHICS 2015. First approach able to synthesize the extrinsic geometry of a shape from intrinsic information. An early version of this work was published as the technical report arXiv:1406.1925 on June 2014*

**Coulomb shapes: Using electrostatic forces for deformation-invariant shape representation**

EUROGRAPHICS-W 2014

D. Boscaini, R. Girdziusas, M.M. Bronstein  
*Oral presentation at 3DOR 2014. Presented also as a poster at the International Computer Vision Summer School (ICVSS), 2014*

**A sparse coding approach for local-to-global 3D shape description**

The Visual Computer, 2014

D. Boscaini, U. Castellani  
*Invited paper. Journal extension of the 3DOR 2013 conference paper*

**Local signatures quantization by sparse coding**

EUROGRAPHICS-W 2013

D. Boscaini, U. Castellani  
*Oral presentation at 3DOR 2013. Presented also as a poster at SGP 2013*

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**Patents**

**US patent application No. 17675011**

Clustered dynamic graph convolutional neural network for biometric 3D hand recognition  
*Inventors: J. Svoboda, P. Astolfi, D. Boscaini, J. Masci*

**US patent No. 10210430**

Filed Feb. 19, 2019

System and a method for learning features on geometric domains (CIP)  
*Inventors: M.M. Bronstein, D. Boscaini, F. Monti • Acquired by Twitter Inc.*

**US patent No. 10013653**

Filed Jul. 3, 2018

System and a method for learning features on geometric domains  
*Inventors: M.M. Bronstein, D. Boscaini, J. Masci, P. Vanderghelynst • Acquired by Twitter Inc.*

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**Invited talks**

**Object 6D pose estimation with vision and language**

Dec. 18, 2025

University of Catania, Italy • Invited by Francesco Ragusa, Giovanni Farinella

**3D object understanding on the shoulders of 2D foundation models**

Jul. 29, 2025

Shandong University, China • Invited by Fabio Poiesi

**Object 6D pose estimation in the foundation models era**

Nov. 22, 2024

University of Trento, Italy • Master course in “Trends and Applications in Computer Vision”, Invited by Massimiliano Mancini

**Object 6D pose estimation in the foundation models era**

Jun. 6, 2024

Politecnico di Torino, Italy • Invited by Francesca Pistilli

**3D object understanding on the shoulders of 2D foundation models**

Mar. 28, 2024

École Polytechnique, Paris, France • Invited by Maks Ovsjanikov

**3D deep learning to the test of real-world challenges**

Dec. 11, 2020

University of Pisa, Italy • Ph.D. event “Visions of Tomorrow”

**3D Deep Learning**

Dec. 11, 2019

Politecnico di Torino, Italy • Invited by Tatiana Tommasi

**Geometric deep learning for 3D shape analysis**

May 13, 2019

Politecnico di Torino, Italy • Invited by Barbara Caputo

**Geometric deep learning for shape analysis**

Sep. 2, 2017

Kos, Greece • Keynote at EUSIPCO 2017

**Geometric deep learning for shape analysis**

Apr. 4, 2017

FBK-TeV, Trento, Italy • Invited by Samuel Rota Bulò and Stefano Messelodi

|                                                                                                                                    |               |
|------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Geometric deep learning for shape analysis<br>CNR-IMATI, Genoa, Italy • Invited by Michela Spagnuolo                               | Feb. 13, 2017 |
| Deep learning on geometric data<br>Zurich, Switzerland • SSSTC RiC big data research workshop                                      | Feb. 16, 2016 |
| Deep learning on geometric data<br>Embedded Vision Systems (eVS), Verona, Italy • Invited by Roberto Marzotto                      | Feb. 8, 2016  |
| Deep learning on geometric data<br>Rainbow group, University of Cambridge, UK • Invited by Flora Tasse                             | Feb. 4, 2016  |
| Deep learning on geometric data<br>University of Cambridge, UK • C.A.K.E. seminar, Invited by Simone Parisotto                     | Feb. 3, 2016  |
| Convolutional neural networks on non-Euclidean domains<br>Potsdam, Germany • Keynote at SciCADE 2015                               | Sep. 14, 2015 |
| Shape-from-operators: recovering shapes from intrinsic differential operators<br>TUM, Munich, Germany • Invited by Emanuele Rodolà | Nov. 26, 2014 |
| Shape-from-operators: recovering shapes from intrinsic differential operators<br>Disentis, Switzerland • USI-ICS retreat           | Aug. 19, 2014 |

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## Teaching experience

### Academic courses

|                                            |                                 |
|--------------------------------------------|---------------------------------|
| Trends and Applications in Computer Vision | University of Trento, Fall 2023 |
|--------------------------------------------|---------------------------------|

### Short courses and tutorials

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|---------------------------------------------------------------------------------------|--------------------|
| Functional Maps: A Flexible Representation for Learning and Computing Correspondences | 3DV 2018           |
| Geometric Deep Learning                                                               | SIGGRAPH Asia 2016 |
| Deep Learning for Shape Analysis                                                      | EUROGRAPHICS 2016  |

### Teaching Assistantships

|                                                  |                                                 |
|--------------------------------------------------|-------------------------------------------------|
| Computer Vision and Pattern Recognition          | Università della Svizzera italiana, Spring 2017 |
| Computer Vision and Pattern Recognition          | Università della Svizzera italiana, Spring 2016 |
| Large Scale Optimization                         | Università della Svizzera italiana, Spring 2016 |
| Computer Graphics                                | Università della Svizzera italiana, Fall 2014   |
| Geometric Image Processing and Computer Vision   | Università della Svizzera italiana, Spring 2014 |
| Calculus                                         | Università della Svizzera italiana, Fall 2013   |
| Mathematical Analysis 1, Mathematical Analysis 2 | University of Verona, 2012–2013                 |
| Mathematical Analysis 1, Mathematical Analysis 2 | University of Verona, 2011–2012                 |
| Mathematical Analysis 1                          | University of Verona, 2010–2011                 |

### Student supervision

|                                                                                                                                                                                        |                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Jaime Corsetti, PhD student at FBK and University of Trento<br>Project: Vision-language models for embodied AI                                                                         | Nov. 2023–present   |
| Mattia Nardon, Master student at University of Trento<br>Role: Internship and Master thesis advisor<br>Project: AI-powered visual monitoring of Lego assembly tasks                    | Mar.–Dec. 2024      |
| Alice Fasoli, Master student at University of Trento<br>Role: Internship and Master thesis advisor<br>Project: Retrieval-driven 6D pose estimation of unseen objects                   | Mar.–Dec. 2024      |
| Matteo Minardi, Master student at University of Trento<br>Role: Internship and Master thesis advisor<br>Projects: Eye-gaze estimation using smart glasses, Vision encoder role in VLMs | Mar.–Oct. 2024      |
| Marco Garosi, Master student at University of Trento<br>Project: Zero-shot semantic segmentation of 3D objects<br>Outcome: Publication at WACV 2025                                    | Dec. 2023–Apr. 2024 |
| Riccardo Tedoldi, Master student at University of Trento                                                                                                                               | Dec. 2023–Apr. 2024 |

|                                                                                    |                |
|------------------------------------------------------------------------------------|----------------|
| Project: Zero-shot semantic segmentation of 3D objects                             |                |
| Outcome: Publication at WACV 2025                                                  |                |
| Jaime Corsetti, Master student at University of Trento                             | 2022–Oct. 2023 |
| Projects: Open-vocabulary and Supervised object 6D pose estimation for RGBD images |                |
| Outcome: Publication at CVPR 2024                                                  |                |
| Safa Abbes, Master student at University of Trento                                 | 2022–2023      |
| Role: Masther thesis coadvisor                                                     |                |
| Project: Self-supervised domain adaptation for RGB images                          |                |
| Antonio Alliegro, PhD student at Politecnico di Torino                             | 2020–2021      |
| Project: Self-Supervised domain adaptation for 3D point clouds                     |                |
| Outcome: Publications at ECCV-W 2020 and ICPR 2021                                 |                |
| Pietro Astolfi, PhD student at FBK, UniTN, and IIT                                 | 2019–2021      |
| Project: Geometric Deep Learning for brain structure analysis                      |                |
| Outcome: Publications at MICCAI 2020, IJCB 2020, and MIA 2023                      |                |
| Levi O. Vasconcelos, PhD student at UniTN and IIT                                  | 2019–2020      |
| Project: Structured domain adaptation                                              |                |
| Outcome: Publications at 3DV 2019 and ICPR 2020                                    |                |
| Antonio Alliegro, Master student at Politecnico di Torino                          | 2019–2020      |
| Role: Masther thesis coadvisor                                                     |                |
| Piero Cavalcanti, Master student at Politecnico di Torino                          | 2019–2020      |
| Role: Masther thesis coadvisor                                                     |                |
| Myriam Bronstein, Master student at Università della Svizzera italiana             | 2016           |
| Project: Machine learning methods on manifolds and graphs                          |                |
| Fatemeh Chegini, Master student at Università della Svizzera italiana              | 2014–2015      |
| Project: Spectral methods for cross-modal retrieval                                |                |

## Academic service

### Conferences revision activity

|                                                                                    |                        |
|------------------------------------------------------------------------------------|------------------------|
| International Conference on Robotics and Automation (ICRA)                         | 2022, 2020             |
| International Conference on Pattern Recognition (ICPR)                             | 2022, 2020             |
| Symmetry and Geometry in Neural Representations (NeurIPS Workshops)                | 2022                   |
| International Conference on Image Analysis and Processing (ICIAP)                  | 2022                   |
| International Conference on Machine Learning, Optimization, and Data Science (LOD) | 2022                   |
| Symposium On Applied Computing (SAC)                                               | 2022                   |
| International Conference on 3D Vision (3DV)                                        | 2021, 2020, 2019, 2018 |
| International Conference on Machine Learning, Optimization, and Data Science (LOD) | 2021                   |
| International Conference on Machine Vision Applications (MVA)                      | 2021, 2019             |
| EUROGRAPHICS                                                                       | 2019, 2017, 2015       |
| The British Machine Vision Conference (BMVC)                                       | 2018                   |
| Computer Vision and Pattern Recognition (CVPR)                                     | 2017                   |
| International Symposium on Vision, Modeling and Visualization (VMV)                | 2016                   |
| Neural Information Processing Systems (NeurIPS)                                    | 2016                   |

### Journal revision activity

|                                                                            |                  |
|----------------------------------------------------------------------------|------------------|
| Robotics and Automation Letters (RAL)                                      | 2022             |
| Computer Graphics Forum (CGF)                                              | 2022             |
| IEEE Transactions on Image Processing (TIP)                                | 2022, 2021       |
| IEEE Transactions on Transactions on Knowledge and Data Engineering (TKDE) | 2022, 2021       |
| Neural Processing Letters (NEPL)                                           | 2022             |
| IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI)      | 2021, 2020       |
| IEEE Transactions on Visualization and Computer Graphics (TVCG)            | 2020, 2018, 2017 |
| Computers and Graphics                                                     | 2019             |

|                                                                  |                  |
|------------------------------------------------------------------|------------------|
| Computer Vision and Image Understanding (CVIU)                   | 2019, 2015       |
| International Journal of Machine Learning and Cybernetics (JMCL) | 2019             |
| Pattern Recognition Letters                                      | 2019             |
| The Visual Computer Journal (TVCJ)                               | 2018, 2017, 2016 |
| Computer Aided Geometric Design (CAGD)                           | 2018             |
| Computer-Aided Design (CAD)                                      | 2018             |
| Sensors                                                          | 2018             |
| IPSJ Transactions on Computer Vision and Applications            | 2017             |

#### **Area chair**

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|------------------------------------------|------|
| British Machine Vision Conference (BMVC) | 2025 |
| British Machine Vision Conference (BMVC) | 2024 |

#### **Program committee**

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|-------------------------------------------------------------------------------------|------|
| International Workshop on Advances in Drone Vision (ADV)                            | 2025 |
| Graph Models for Learning and Recognition (GMLR)                                    | 2022 |
| Organized within the 37th ACM Symposium on Applied Computing, Brno (Czech Republic) |      |

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